TEXORA · LECTURE SHEET

Pigment Printing

Half the world's printing: a binder film instead of a dye bond — formulation, curing, special effects, and the troubleshooting of rub fastness and handle.

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 10 · LEVEL 3 TERM 1

1. Why pigment is the printer's default

About half of all textile printing worldwide is pigment printing. The reasons mirror Topic 4: one paste system serves every fibre and blend, there is **no wash-off** (the flow chart of Topic 8 loses step 6), the print is visible and correctable immediately, and light fastness is excellent. The costs are the familiar ones — surface colour means rub fastness and handle must be engineered, and metallic/white effects depend entirely on pigment choice.

2. The pigment print paste

COMPONENT	ROLE AND TYPICAL LEVEL (PER KG PASTE)
Pigment dispersion	Colour; 10–80 g/kg by depth (white prints: 150–300 g/kg titanium-dioxide-based white)
Binder	Self-crosslinking acrylic copolymer; 100–250 g/kg — enough to bury the pigment load fully (rule: binder solids ≥ pigment solids for deep shades)
Thickener	Synthetic polyacrylate emulsion thickener, 15–30 g/kg, or emulsion (oil-in-water) systems in legacy plants
Crosslinker / fixing agent	Melamine-formaldehyde resin 10–20 g/kg with catalyst (e.g. magnesium chloride or ammonium salts) to raise wash/rub grades
Softener	Silicone emulsion 10–20 g/kg restoring surface slip and hand
Auxiliaries	Ammonia to pH 8–9, defoamer, urea small amounts as humectant, water to weight

3. Print – dry – cure: the fixation that is only a film

After printing, the fabric is dried (100–110 °C) and **cured at 140–160 °C for 2–5 min**. During curing the binder coalesces into a continuous film and crosslinks — with itself, with the crosslinker, and (for reactive-site binders) partly to fibre hydroxyls. Fastness is born here and nowhere else: an under-cured pigment print fails rub and wash tests no matter how good the paste was; over-curing wastes energy, yellows whites and can release formaldehyde from resin crosslinkers (low-formaldehyde and formaldehyde-free systems are now buyer standard). Because there is no steaming or washing, pigment printing has the lowest water footprint of any style — the environmental argument that keeps expanding its share.

4. Special pigment effects

- **White and opaque colour on dark grounds:** titanium dioxide white as base or in-paste opacifier — impossible with transparent dyes.
- **Metallic:** aluminium or bronze powder pastes with binder (tarnish and rub are the limits).
- **Puff (expanding) prints:** heat-expanding microspheres raise the design in relief during curing.
- **Glitter, foil, flock:** adhesive printed as the design, decoration applied to it (Topic 9).
- **Discharge-style pigment:** pigment printed on deep grounds gives a discharge look without discharge chemistry, at some cost in handle.

5. Troubleshooting table

FAULT	CAUSE AND CURE
Poor dry rub fastness	Under-cure or pigment above binder capacity → raise cure, raise binder/crosslinker ratio
Harsh handle	Excess binder/film thickness → reduce binder to the fastness minimum, add silicone softener, use finer pigment dispersions
Screen blocking	Paste drying in mesh → humectant, print-room humidity, clean screens on schedule (Experiment 6 skills)
Colour loss in washing	Film not crosslinked (catalyst omitted/expired) → verify crosslinker and catalyst, re-prove cure temperature with a temperature strip

6. Glossary — key terms

- **Coalescence:** merging of binder emulsion particles into a continuous film on drying/heating.
- **Opacifier:** high-refractive-index pigment (TiO_2) hiding the ground beneath.
- **Crocking:** colour transfer by rubbing — the everyday name for poor rub fastness.

7. Key points

- Pigment printing = print, dry, cure; no steam, no wash-off.
- Binder-to-pigment ratio and cure are the two fastness variables; handle is their price.
- Opaque whites and effects on dark grounds are pigment's unique territory.

8. Summary

Pigment printing trades the molecular elegance of dye fixation for the engineering simplicity of a cured film. Managed with disciplined binder ratios and verified curing, it delivers economical, blend-proof, wash-free printing — and its limits (rub, handle) are understood quantitatively rather than discovered at inspection.

9. Practice

1. A paste carries 60 g/kg pigment. Using the binder-solids rule, state the minimum binder solids per kg and two faults expected if binder is halved.
2. Why does pigment printing skip wash-off, and what quality risk replaces the wash-off risk of dye printing?
3. A white print on black cotton looks grey. Explain optically and give the paste correction.

Answer key

1. At least 60 g/kg binder solids for a deep shade (binder emulsion as supplied is part water — convert by its solids content). Halving: pigment particles protrude from the film → severe crocking (rub grade falls) and wash loss of colour as surface pigment detaches.
2. There is no unfixed soluble dye or thickener to remove — everything printed is meant to stay as film. The replaced risk is curing risk: fastness depends on a heat step whose failure is invisible until rub/wash testing.
3. The white film is too thin/translucent, so the black ground shows through and greys the white (low hiding power). Increase TiO_2 white pigment load and film thickness (or print a white base layer), checking handle cost.

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